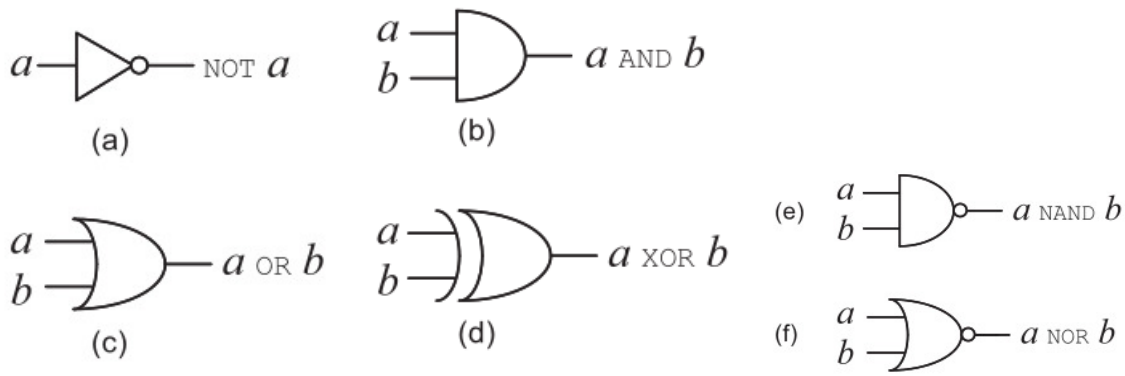
Lecture 5 (section 1.3.7,1.4.1-1.4.4)

Quantum algorithms

Classical computations on quantum computer

- Typical classical gates:



$\Longrightarrow$

$$(a):\ a \xrightarrow{\text{NOT}} a \oplus 1\,, \qquad (b):\ (a,b) \xrightarrow{\text{AND}} a \cdot b\,, \qquad (c):\ (a,b) \xrightarrow{\text{OR}} a \cdot b \oplus a \oplus b$$

$$(d):\ (a,b) \xrightarrow{\text{XOR}} a \oplus b\,, \qquad (e):\ (a,b) \xrightarrow{\text{NAND}} ab \oplus 1 \qquad (f):\ (a,b) \xrightarrow{\text{NOR}} a \cdot b \oplus a \oplus b \oplus 1$$

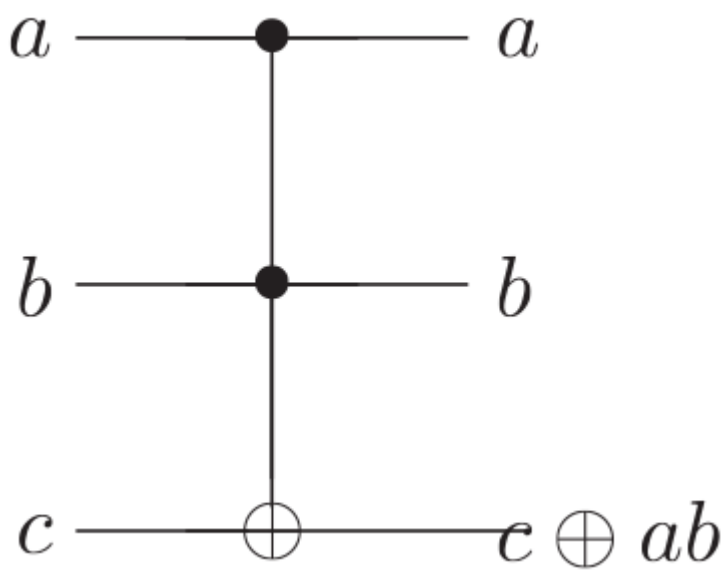
  - NAND gate is **universal**, *e.g.,*

$$\text{NOT} - a \ \equiv a - \text{NAND} - 1$$

  (google for other gates)
  - of the 6 gates only (a) is *reversible*, and the other ones are *irreversible*

- All quantum gates are reversible (because they are unitaries)

- To simulate classical computer we need reversible gate for the above logical operations



- Solution, a *Toffoli* gate, $\mathcal{T}$,

$$\mathcal{T}^2 : (a, b, c) = (a, b, c \oplus ab \oplus ab) = (a, b, c)$$

so $\mathcal{T}^{-1} = \mathcal{T}$

$\Longrightarrow$ note that

$$\mathcal{T}(a, b, 1) = (a, b, 1 \oplus ab) = (a, b, a \text{ NAND } b)$$

i.e., Toffoli gate simulate (in a reversible manner) universal classical gate NAND.

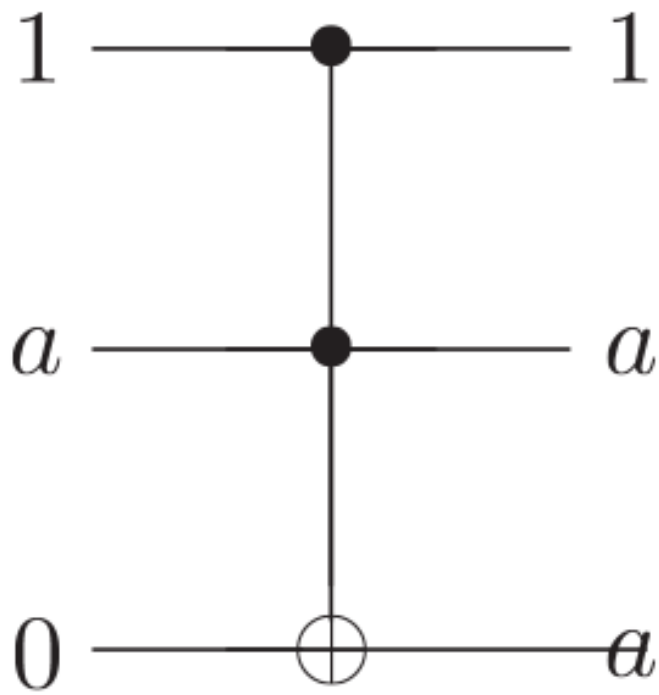
- Toffoli gate is a valid quantum 3-qubit gate:

$$|\psi\rangle = \sum_{ijk} a_{ijk} \cdot |ijk\rangle \ \xrightarrow{\mathcal{T}} \ \sum_{ijk} a_{ijk} \cdot |ij\rangle \otimes |i \cdot j \oplus k\rangle$$

(write a matrix representation for a quantum Toffoli gate, and verify that it is unitary)

- a different between classical and quantum $\mathcal{T}$:
  - classical $\mathcal{T}$ can copy a bit:

$$\mathcal{T} : (1, a, 0) = (1, a, 0 \oplus 1 \cdot a) = (1, a, a)$$



- what would identical quantum gate do? (it should not be able to copy!)

$$|1\rangle \otimes \underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{|\psi\rangle} \otimes |0\rangle = \alpha|100\rangle + \beta|110\rangle \xrightarrow{\mathcal{T}} \alpha|100\rangle + \beta|111\rangle \neq |1\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

as

$$|1\rangle \otimes |\psi\rangle \otimes |\psi\rangle = |1\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha|0\rangle + \beta|1\rangle) = \alpha^2|100\rangle + \alpha\beta|101\rangle + \alpha\beta|110\rangle + \beta^2|111\rangle$$

Quantum teleportation

- The story:
  - Alice and Bob are in different galaxies
  - they share a Bell state $|\beta_{00}\rangle$, where Alice has the first qubit, and Bob the second

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}} \underbrace{|0\rangle}_{\text{Alice's}} \otimes \underbrace{|0\rangle}_{\text{Bob's}} + \frac{1}{\sqrt{2}} \underbrace{|1\rangle}_{\text{Alice's}} \otimes \underbrace{|1\rangle}_{\text{Bob's}}$$

- Alice also has an arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
- The initial state is thus

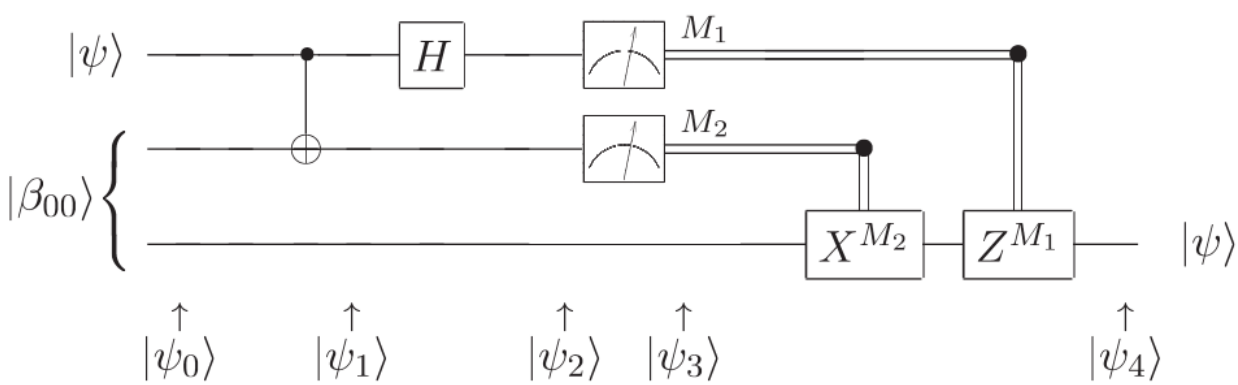
$$\underbrace{|\psi\rangle_0}_{\text{Alice+Bob state}} \equiv |\psi\rangle|\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$= \frac{\alpha}{\sqrt{2}} \cdot \underbrace{|00\rangle}_{\text{Alice's}} \otimes \underbrace{|0\rangle}_{\text{Bob's}} + \frac{\alpha}{\sqrt{2}} \cdot \underbrace{|01\rangle}_{\text{Alice's}} \otimes \underbrace{|1\rangle}_{\text{Bob's}} + \frac{\beta}{\sqrt{2}} \cdot \underbrace{|10\rangle}_{\text{Alice's}} \otimes \underbrace{|0\rangle}_{\text{Bob's}} + \frac{\beta}{\sqrt{2}} \cdot \underbrace{|11\rangle}_{\text{Alice's}} \otimes \underbrace{|1\rangle}_{\text{Bob's}}$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

- Alice and Bob are allowed to do <u>*local*</u> quantum gates (meaning only involving the qubits they own); they also allowed to classically communicate (talk to one other)
- The task: Bob must have exactly $|\psi\rangle$

- The teleportation circuit:



- step-1:

$$|\psi\rangle_0 \;\rightarrow\; |\psi\rangle_1 = (U_{CN} \otimes I)|\psi\rangle_0$$

$$\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle \;\xrightarrow{U_{CN}\otimes I}\; \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|110\rangle + \frac{\beta}{\sqrt{2}}|101\rangle$$

- step-2:

$$|\psi\rangle_1 \;\rightarrow\; |\psi\rangle_2 = (H \otimes I \otimes I)|\psi\rangle_1$$

$$\frac{\alpha}{\sqrt{2}}|000\rangle+\frac{\alpha}{\sqrt{2}}|011\rangle+\frac{\beta}{\sqrt{2}}|110\rangle+\frac{\beta}{\sqrt{2}}|101\rangle \xrightarrow{H\otimes I\otimes I} \frac{\alpha}{\sqrt{2}}\frac{|0\rangle+|1\rangle}{\sqrt{2}}\otimes|00\rangle+\frac{\alpha}{\sqrt{2}}\frac{|0\rangle+|1\rangle}{\sqrt{2}}\otimes|11\rangle$$
$$+\frac{\beta}{\sqrt{2}}\frac{|0\rangle-|1\rangle}{\sqrt{2}}\otimes|10\rangle+\frac{\beta}{\sqrt{2}}\frac{|0\rangle-|1\rangle}{\sqrt{2}}\otimes|01\rangle$$
$$=\frac{\alpha}{2}|000\rangle+\frac{\alpha}{2}|100\rangle+\frac{\alpha}{2}|011\rangle+\frac{\alpha}{2}|111\rangle+\frac{\beta}{2}|010\rangle-\frac{\beta}{2}|110\rangle+\frac{\beta}{2}|001\rangle-\frac{\beta}{2}|101\rangle$$
$$=\frac{1}{2}\Big[|00\rangle\otimes(\alpha|0\rangle+\beta|1\rangle)+|01\rangle\otimes(\alpha|1\rangle+\beta|0\rangle)+|10\rangle\otimes(\alpha|0\rangle-\beta|1\rangle)+|11\rangle\otimes(\alpha|1\rangle-\beta|0\rangle)\Big]$$

- step-3: Alice measures her qubits and obtains the result M_1M_2:

$$|\psi\rangle_3:\ M_1M_2=\begin{cases} 00\,, & \text{probability}=\left|\frac{1}{2}\right|^2=\frac{1}{4}\,, \quad \text{Bob's state}: |Bob\rangle_{00}=\alpha|0\rangle+\beta|1\rangle \\ 01\,, & \text{probability}=\left|\frac{1}{2}\right|^2=\frac{1}{4}\,, \quad \text{Bob's state}: |Bob\rangle_{01}=\alpha|1\rangle+\beta|0\rangle \\ 10\,, & \text{probability}=\left|\frac{1}{2}\right|^2=\frac{1}{4}\,, \quad \text{Bob's state}: |Bob\rangle_{10}=\alpha|0\rangle-\beta|1\rangle \\ 11\,, & \text{probability}=\left|\frac{1}{2}\right|^2=\frac{1}{4}\,, \quad \text{Bob's state}: |Bob\rangle_{11}=\alpha|1\rangle-\beta|0\rangle \end{cases}$$

- step-4: Alice calls Bob and tells a pair M_1M_2; Bob <u>*knows*</u> that he has $|Bob\rangle_{M_1M_2}$ qubit; he performs

$$|\psi\rangle_3 \ \rightarrow\ |\psi\rangle_4^{M_1M_2}=Z^{M_1}\cdot X^{M_2}\cdot|Bob\rangle_{M_1M_2}$$

resulting in

$$|\psi\rangle_4^{M_1M_2}=\begin{cases} 00:\ Z^0X^0|Bob\rangle_{00}=I|Bob\rangle_{00}=|\psi\rangle \\ 01:\ Z^0X^1|Bob\rangle_{01}=X|Bob\rangle_{01}=X(\alpha|1\rangle+\beta|0\rangle)=\alpha|0\rangle+\beta|1\rangle=|\psi\rangle \\ 10:\ Z^1X^0|Bob\rangle_{10}=Z|Bob\rangle_{10}=Z(\alpha|0\rangle-\beta|1\rangle)=\alpha|0\rangle+\beta|1\rangle=|\psi\rangle \\ 11:\ Z^1X^1|Bob\rangle_{11}=ZX|Bob\rangle_{11}=Z\underbrace{X(\alpha|1\rangle-\beta|0\rangle)}_{=\alpha|0\rangle-\beta|1\rangle}=\alpha|0\rangle+\beta|1\rangle=|\psi\rangle \end{cases}$$

$\Longrightarrow$ **teleportation is succesful!**

- NOTE:
  - teleportation does not break special relativity — we will see formally later that before Alice's call Bob does not have any knowledge about $|\psi\rangle$ — the communication via classical phone lines does not break the speed of light limit

▪ teleportation does not break no-cloning — once $|\psi\rangle$ is teleported, Alice does not have it!

Deutsch–Jozsa algorithm

- The story:

  ▪ $f(x)$ is a function with the n-bit domain, *i.e.,* $x = \{0, \cdots 2^n - 1\}$ and a single-bit range $\{1, 0\}$. We know that $f(x)$ is either a constant function (the same for all x), or balanced (exactly $\frac{1}{2}$ numbers from the domain evaluates to 0, and the other half evaluates to 1). How many evaluations of f are necessary to determine (for sure) whether f is a constant or balanced?

  ▪ classical computations:

$$\frac{2^n}{2} + 1$$

  evaluations of f

  ▪ quantum computations:

$$1$$

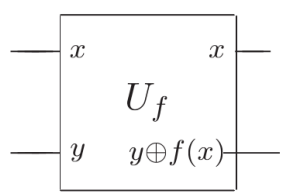
  evaluation of f

- Some notations for quantum circuit:

  ▪ we write x in a bit notation, and encode it as an n-qubit state, *e.g.,* for $n = 4$ and $x = 13$

$$x = 1 \cdot 2^0 + 0 \cdot 2^1 + 1 \cdot 2^2 + 1 \cdot 2^3 = 1 + 0 + 2 + 8 = 13 \qquad \Longleftrightarrow \qquad |x\rangle = |1011\rangle$$

  ▪ $(n+1)$-qubit unitary U_f that uses classical evaluation $f(x)$:



$$|xy\rangle \xrightarrow{U_f} |x\rangle \otimes |y \oplus f(x)\rangle$$

$\Longrightarrow$ why U_f is unitary?

Lets check for $n = 1$:

- $|xy\rangle$ are 2-qubit states
- evaluation on the basis set:

$$|00\rangle \xrightarrow{U_f} |0\rangle \otimes |f(0)\rangle$$
$$|01\rangle \xrightarrow{U_f} |0\rangle \otimes |\overline{f(0)}\rangle$$
$$|10\rangle \xrightarrow{U_f} |1\rangle \otimes |f(1)\rangle$$
$$|11\rangle \xrightarrow{U_f} |1\rangle \otimes |\overline{f(1)}\rangle$$

$\Longrightarrow$ the 4×4 matrix representation of the transformation is

$$U_f = \begin{bmatrix} UP & 0_{2\times 2} \\ 0_{2\times 2} & DOWN \end{bmatrix}$$

where

$$UP = \begin{cases} I\,, & \text{if } f(0) = 0 \\ X\,, & \text{if } f(0) = 1 \end{cases}, \qquad DOWN = \begin{cases} I\,, & \text{if } f(1) = 0 \\ X\,, & \text{if } f(1) = 1 \end{cases}$$

Since UP and $DOWN$ blocks are unitary, U_f is unitary

- Note that

$$H^{\otimes 2}|00\rangle = H|0\rangle \otimes H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2^2}}(|00\rangle + |10\rangle + |01\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2^2}}(\underbrace{|0\rangle}_{=00} + |\underbrace{1}_{=10}\rangle + |\underbrace{2}_{=01}\rangle + |\underbrace{3}_{=11}\rangle) = \frac{1}{\sqrt{2^2}} \sum_{x=0}^{2^2-1} |x\rangle$$

similarly:

$$H^{\otimes n}|0\rangle^{\otimes n} = (H|0\rangle)^{\otimes n} = \frac{1}{\sqrt{2^n}} \cdot \underbrace{(|0\rangle + |1\rangle) \otimes \cdots \otimes (|0\rangle + |1\rangle)}_{n-\text{times}} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

- Let's compute

$$U_f\,|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\left(|x\rangle \otimes |f(x)\rangle - |x\rangle \otimes |\overline{f(x)}\rangle\right) = \begin{cases} \frac{1}{\sqrt{2}}(|x\rangle \otimes |0\rangle - |x\rangle \otimes |1\rangle)\,, & f(x) = 0 \\ \frac{1}{\sqrt{2}}(|x\rangle \otimes |1\rangle - |x\rangle \otimes |0\rangle)\,, & f(x) = 1 \end{cases}$$

$$= (-1)^{f(x)} \cdot \frac{1}{\sqrt{2}}(|x\rangle \otimes |0\rangle - |x\rangle \otimes |1\rangle) = (-1)^{f(x)}|x\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

- Notice the following

$$H|y\rangle = \frac{1}{\sqrt{2}}\sum_{z=0}^{1}(-1)^{y\cdot z}|z\rangle$$

Indeed:

$$H|0\rangle = \frac{1}{\sqrt{2}}\Big((-1)^{0\cdot 0}|0\rangle + (-1)^{0\cdot 1}|1\rangle\Big) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}\Big((-1)^{1\cdot 0}|0\rangle + (-1)^{1\cdot 1}|1\rangle\Big) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Similarly:

$$H^{\otimes 2}|y_1y_2\rangle = H|y_1\rangle \otimes H|y_2\rangle = \frac{1}{\sqrt{2}}\sum_{z_1=0}^{1}(-1)^{y_1\cdot z_1}|z_1\rangle \otimes \frac{1}{\sqrt{2}}\sum_{z_2=0}^{1}(-1)^{y_2\cdot z_2}|z_2\rangle$$

$$= \frac{1}{\sqrt{2^2}}\sum_{z_1}\sum_{z_2}(-1)^{y_1\cdot z_1 + y_2\cdot z_2}|z_1z_2\rangle$$

and more generally:

$$H^{\otimes n}|x\rangle = H^{\otimes n}|x_1\cdots x_n\rangle = (H|x_i\rangle)^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_{z_1}\cdots\sum_{z_n}(-1)^{x_1\cdot z_1+\cdots+x_n\cdot z_n}|z_1\cdots z_n\rangle$$

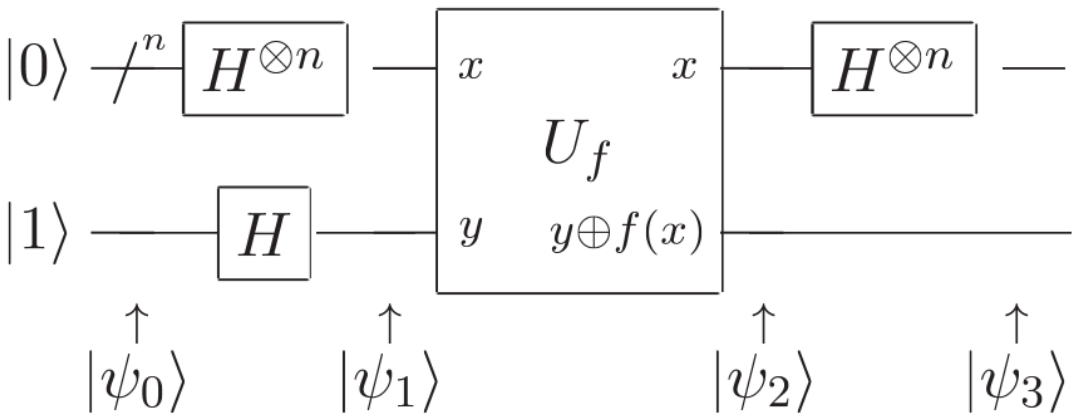
$$= \frac{\sum_z(-1)^{[x\cdot z]}|z\rangle}{\sqrt{2^n}}$$

where $[x\cdot z]$ is a bitwise inner product modulo 2, *i.e.*,

$$[7\cdot 23] = \left[\underbrace{111}_{=1\cdot 2^0+1\cdot 2^1+1\cdot 2^2=1+2+4} \cdot \underbrace{11101}_{=1\cdot 2^0+1\cdot 2^1+1\cdot 2^2+0\cdot 2^3+1\cdot 2^4=1+2+4+0+16}\right] = [11100\cdot 11101]$$

$$= 1+1+1+0+0 = 1$$

- Deutsch-Jozsa algorithm:



- step-1:

$$|\psi\rangle_0 = (|0\rangle)^{\otimes n} \otimes |1\rangle$$

- step-2:

$$|\psi\rangle_1 = H^{\otimes n} \otimes H \ |\psi\rangle_0 = H^{\otimes n} \left(|0\rangle\right)^{\otimes n} \otimes H|1\rangle = \sum_{x=0}^{2^n-1} \frac{|x\rangle}{\sqrt{2^n}} \otimes \ \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

- step-3:

$$|\psi\rangle_2 = U_f|\psi\rangle_1 = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} \ U_f|x\rangle \otimes \ \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right] = \sum_{x=0}^{2^n-1} \frac{(-1)^{f(x)}|x\rangle}{\sqrt{2^n}} \otimes \ \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

- step-4:

$$|\psi\rangle_3 = H^{\otimes n} \otimes I|\psi\rangle_2 = \sum_x \frac{(-1)^{f(x)}}{\sqrt{2^n}} \ H^{\otimes n}|x\rangle \otimes I \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right] = \sum_x \sum_z \frac{(-1)^{[x\cdot z]+f(x)}|z\rangle}{2^n} \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right]$$

- Let's analyze the amplitude of $|0\rangle^{\otimes n} \otimes |-\rangle$ in $|\psi\rangle_3$:

$$\psi_3 = \sum_x \frac{(-1)^{[x\cdot 0]+f(x)}}{2^n} \ |0\rangle^{\otimes n} \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right] + \text{orthogonal}$$

$\Longrightarrow$

$$\sum_x \frac{(-1)^{[x\cdot 0]+f(x)}}{2^n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} = \begin{cases} 1\,, & \text{if } f(x) = \text{const} \\ 0\,, & \text{if } f(x) = \text{balance} \end{cases}$$

▪ step-5: measure the first-n qubits; if the result of the measurement is anything other than $|0\rangle^{\otimes n}$, the function $f(x)$ is **balanced**, otherwise, it is **constant**

General comments on quantum algorithms:

- Deutsch–Jozsa algorithm is exponentially more efficient than the corresponding classical algorithm (as $n \to \infty$)
- Deutsch–Jozsa algorithm does not have practical applications
- All quantum algorithms in real world require *quantum error correction*: β_{00} state shared by Alice and Bob in quantum teleportation will not last long.....
- Shor's factoring algorithm (later in the course) is exponentially more efficient, and is useful (?) in breaking RSA encryption (google it)
- Quantum search algorithm (later in the course) provides $\sqrt{}$ speed-up
- Quantum simulation (universal quantum computer can simulate quantum materials exponentially more efficient than classical computers)